ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **Number Play** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 1234	End with the smallest even digit, then arrange the rest in increasing order.
2	**(c)** 90	First digit 1-9 and middle digit 0-9, the last digit is fixed: 9×10.
3	**(c)** 29992	Fill the middle digits with 9.
4	**(a)** 4	Count each step.
5	**(c)** 1441	Keep the first half and mirror it; if that is not bigger, increase the middle.
6	**(a)** 465	Pair the numbers: first with last.
7	**(b)** 6174	Largest - smallest.
8	**(d)** 9100	Put 9s from the left until the sum is used up.
9	**(c)** 2576	End with the smallest even digit, then arrange the rest in increasing order.
10	**(c)** 49994	Fill the middle digits with 9.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Greatest 97431 (1); Smallest 13479 (1).	1+1
12	87, 165, 726, 1353, 4884 (1); Palindrome 4884 (1).	1+1
13	Second row: 5, 4, 5; Third row: 9, 9 (1); Top $= 18$ (1).	1+1
14	Greatest 86520 (1); Smallest 20568 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Step 1: 8172 (1); Step 2: 7443 (1); Digits used: arrange 8172 to get the next number (1).	1+1+1
16	Largest below: 1221 (1); Smallest above: 1331 (1); Check by reversing (1).	1+1+1
17	Pairs sum to 41 (1); Number of pairs $= 40 \div 2 = 20$ (1); Sum $= 20 \times 41 = 820$ (1).	1+1+1
	OR: Step 1: 6264 (1); Step 2: 4176 (1); Digits used: arrange 6264 to get the next number (1).	1+1+1
18	Largest below: 2442 (1); Smallest above: 2552 (1); Check by reversing (1).	1+1+1
	OR: Pairs sum to 61 (1); Number of pairs $= 60 \div 2 = 30$ (1); Sum $= 30 \times 61 = 1830$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	The last digit is also 2, and the middle two digits are equal: $2 \times 2 + 2m = 10$ so $m = 3$ (1); Number 2332 (1); Reverse is the same: palindrome (1); Add 11: 2343 (1); Not a palindrome (1).	1+1+1+1+1
	OR: Try arrangements (1); Closest number is 5137 (2); Difference $= 137$ (2).	1+2+2
20	The last digit is also 3, and the middle two digits are equal: $2 \times 3 + 2m = 14$ so $m = 4$ (1); Number 3443 (1); Reverse is the same: palindrome (1); Add 11: 3454 (1); Not a palindrome (1).	1+1+1+1+1
	OR: Try arrangements (1); Closest number is 6248 (2); Difference $= 248$ (2).	1+2+2

Mistakes to avoid:

- Writing 0 first when forming the smallest number (e.g. 046); the smallest 3-digit number from 4, 0, 6 is 406.
- In a Kaprekar step, subtracting the larger from the smaller or adding the two arrangements.
- Applying the odd rule ($3n + 1$) to an even number or halving an odd number in the hailstone game.
- Reversing digits wrongly in reverse-and-add (the reverse of 68 is 86) or stopping one step early.
- Counting the starting number as a step when counting steps to reach 1 or a palindrome.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **Number Play** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 7	Count each step.
2	**(a)** 4884	Keep the first half and mirror it; if that is not bigger, increase the middle.
3	**(c)** 1035	Pair the numbers: first with last.
4	**(b)** 8532	Largest - smallest.
5	**(c)** 9600	Put 9s from the left until the sum is used up.
6	**(b)** 1398	End with the smallest even digit, then arrange the rest in increasing order.
7	**(d)** 69996	Fill the middle digits with 9.
8	**(c)** 10	Count each step.
9	**(b)** 3003	Keep the first half and mirror it; if that is not bigger, increase the middle.
10	**(c)** 1830	Pair the numbers: first with last.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	96, 165, 726, 1353, 4884 (1); Palindrome 4884 (1).	1+1
12	Second row: 6, 3, 5; Third row: 9, 8 (1); Top $= 17$ (1).	1+1
13	Greatest 97531 (1); Smallest 13579 (1).	1+1
14	175, 746, 1393, 5324, 9559 (1); Palindrome 9559 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Step 1: 6174 (1); Step 2: 6174 (1); Digits used: arrange 6174 to get the next number (1).	1+1+1
16	Largest below: 3773 (1); Smallest above: 3883 (1); Check by reversing (1).	1+1+1
17	Pairs sum to 81 (1); Number of pairs $= 80 \div 2 = 40$ (1); Sum $= 40 \times 81 = 3240$ (1).	1+1+1
	OR: Step 1: 9171 (1); Step 2: 8532 (1); Digits used: arrange 9171 to get the next number (1).	1+1+1
18	Largest below: 5005 (1); Smallest above: 5115 (1); Check by reversing (1).	1+1+1
	OR: Pairs sum to 121 (1); Number of pairs $= 120 \div 2 = 60$ (1); Sum $= 60 \times 121 = 7260$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	The last digit is also 4, and the middle two digits are equal: $2 \times 4 + 2m = 10$ so $m = 1$ (1); Number 4114 (1); Reverse is the same: palindrome (1); Add 11: 4125 (1); Not a palindrome (1).	1+1+1+1+1
	OR: Try arrangements (1); Closest number is 4619 (2); Difference $= 119$ (2).	1+2+2
20	The last digit is also 5, and the middle two digits are equal: $2 \times 5 + 2m = 22$ so $m = 6$ (1); Number 5665 (1); Reverse is the same: palindrome (1); Add 11: 5676 (1); Not a palindrome (1).	1+1+1+1+1
	OR: Try arrangements (1); Closest number is 2873 (2); Difference $= 127$ (2).	1+2+2

Mistakes to avoid:

- Writing 0 first when forming the smallest number (e.g. 046); the smallest 3-digit number from 4, 0, 6 is 406.
- In a Kaprekar step, subtracting the larger from the smaller or adding the two arrangements.
- Applying the odd rule ($3n + 1$) to an even number or halving an odd number in the hailstone game.
- Reversing digits wrongly in reverse-and-add (the reverse of 68 is 86) or stopping one step early.
- Counting the starting number as a step when counting steps to reach 1 or a palindrome.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **Number Play** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 8352	Largest - smallest.
2	**(d)** 9920	Put 9s from the left until the sum is used up.
3	**(c)** 3456	End with the smallest even digit, then arrange the rest in increasing order.
4	**(a)** 89998	Fill the middle digits with 9.
5	**(a)** 7	Count each step.
6	**(d)** 7667	Keep the first half and mirror it; if that is not bigger, increase the middle.
7	**(a)** 5050	Pair the numbers: first with last.
8	**(b)** 8172	Largest - smallest.
9	**(c)** 9970	Put 9s from the left until the sum is used up.
10	**(c)** 9993	Put 9s from the left until the sum is used up.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Second row: 6, 8, 7; Third row: 14, 15 (1); Top $= 29$ (1).	1+1
12	Greatest 86420 (1); Smallest 20468 (1).	1+1
13	192, 483, 867, 1635, 6996 (1); Palindrome 6996 (1).	1+1
14	Second row: 8, 9, 6; Third row: 17, 15 (1); Top $= 32$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Step 1: 6354 (1); Step 2: 3087 (1); Digits used: arrange 6354 to get the next number (1).	1+1+1
16	Largest below: 6666 (1); Smallest above: 6776 (1); Check by reversing (1).	1+1+1
17	Pairs sum to 151 (1); Number of pairs $= 150 \div 2 = 75$ (1); Sum $= 75 \times 151 = 11325$ (1).	1+1+1
	OR: Step 1: 8352 (1); Step 2: 6174 (1); Digits used: arrange 8352 to get the next number (1).	1+1+1
18	Largest below: 8118 (1); Smallest above: 8228 (1); Check by reversing (1).	1+1+1
	OR: Pairs sum to 201 (1); Number of pairs $= 200 \div 2 = 100$ (1); Sum $= 100 \times 201 = 20100$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	The last digit is also 6, and the middle two digits are equal: $2 \times 6 + 2m = 16$ so $m = 2$ (1); Number 6226 (1); Reverse is the same: palindrome (1); Add 11: 6237 (1); Not a palindrome (1).	1+1+1+1+1
	OR: Try arrangements (1); Closest number is 6952 (2); Difference $= 48$ (2).	1+2+2
20	The last digit is also 7, and the middle two digits are equal: $2 \times 7 + 2m = 24$ so $m = 5$ (1); Number 7557 (1); Reverse is the same: palindrome (1); Add 11: 7568 (1); Not a palindrome (1).	1+1+1+1+1
	OR: Try arrangements (1); Closest number is 1874 (2); Difference $= 126$ (2).	1+2+2

Mistakes to avoid:

- Writing 0 first when forming the smallest number (e.g. 046); the smallest 3-digit number from 4, 0, 6 is 406.
- In a Kaprekar step, subtracting the larger from the smaller or adding the two arrangements.
- Applying the odd rule ($3n + 1$) to an even number or halving an odd number in the hailstone game.
- Reversing digits wrongly in reverse-and-add (the reverse of 68 is 86) or stopping one step early.
- Counting the starting number as a step when counting steps to reach 1 or a palindrome.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.